FIELD CAMPAIGN NK-C25-14

NK-600: expansion valve EXV-2 rev 2 can stick partly closed after idle periods

Bulletin	Category	Issued
SB-2025-014	**Reliability, field campaign**	**2 September 2025**
Campaign code	Product	Revision
NK-C25-14	**NK-600 air-cooled chiller, R-513A**	**0**

Before you leak test or add refrigerant

On an NK-600 in the serial range below with an L-03 alarm, check the EXV position in the service menu first (section 4). A sticking expansion valve starves the evaporator and gives the same alarm as a low charge. Adding refrigerant does not correct it, and the circuit is overcharged when the valve frees again.

AFFECTED UNITS

Model	Serial numbers	Market	Valve fitted	Status
NK-600	NK600-220001 to NK600-231999	EU deliveries	EXV-2 rev 2 (63-1170)	In campaign NK-C25-14
NK-600	NK600-220001 to NK600-231999	Outside EU	Other valve supplier	Not affected
NK-600	NK600-232000 and later	All	EXV-2 rev 3 (63-1180)	Not affected
NK-300	All	All	EXV-1	Not affected

The list of serial numbers in the campaign, with completion status, is available to authorised partners in the Nordkyl partner portal under Campaigns. The rev 2 valve can also be recognised on the unit: its stepper motor has a grey cable with a 4-pin connector; the rev 3 motor has a black cable with a 6-pin connector.

SYMPTOM

- Alarm L-03 (low suction pressure) on one circuit only. The other circuit runs normally.
- Typically in the first hours after a restart that follows an idle period of more than 48 hours: weekend shutdown, low load in winter, or after maintenance. More frequent at low ambient temperature.
- Refrigerant charge is normal: liquid line sight glass clear, subcooling in the normal range, no leak found.
- Suction superheat on the affected circuit is high, while the controller log shows the EXV request rising toward 100%.
- The alarm often clears after a reset and a restart and comes back days or weeks later. This is why units have been topped up in the field without a leak being found.

ROOT CAUSE

The needle of the EXV-2 rev 2 runs in a PTFE guide bushing. During long idle periods at low temperature, a thin film from the POE compressor oil forms between needle and bushing. The stepper motor of rev 2 runs with a low holding current, and at start it does not always have the torque to break the needle free. The motor loses steps and the needle stays at a small opening while the controller asks for more.

The position feedback of the valve reports the real needle position, so the condition can be seen in the service menu. EXV-2 rev 3 (63-1180) has a coated needle without the PTFE bushing and a stronger stepper motor. It has not shown the fault in 18 months of field operation or in cold-soak tests at Nordkyl Aarhus.

DIAGNOSIS: CHECK THE EXV BEFORE LEAK TESTING OR CHARGING

1. Confirm that the unit serial number is in the range above.
2. Let the alarmed circuit run, or restart it, for at least 10 minutes.
3. On the controller, open Service > Circuits > Circuit 1 (or 2) > EXV. Read "EXV request" and "EXV position" in %. Service level login is needed.
4. If the position stays below 15% while the request is higher (usually 40 to 100%), the valve is stuck. The fault is confirmed. Do not add refrigerant. Replace the valves as described below.
5. If the position follows the request within about 5%, the EXV is not the cause of this alarm. Continue with the L-03 troubleshooting in the NK-600 Service Manual. The unit is still in the campaign; schedule the valve replacement.
6. Do not tap the valve body to free it. A valve that moves after tapping is still defective and will stick again.

Write down the request and position values for both circuits. They are needed for the campaign claim.

CORRECTIVE ACTION: REPLACE EXV-2 ON BOTH CIRCUITS

Replace the valve on both circuits of the unit, also when only one circuit has alarmed. The work opens the refrigerant circuit and may only be done by refrigerant-certified technicians (EU F-gas Category I or national equivalent). Allow half a day on site for the unit.

1. Order two valve assemblies 63-1180 (rev 3) and two filter drier cores 63-0410 per unit. Each 63-1180 kit includes the new stepper cable and a campaign label.
2. Stop the unit, isolate it electrically and lock it out.
3. Recover the refrigerant from circuit 1 into a recovery cylinder. Weigh and record the amount recovered.
4. Remove the EXV-2 rev 2 (63-1170). Braze the new valve 63-1180 in place with a nitrogen purge through the circuit. Wrap the valve body in a wet cloth; the body must not exceed 120 C.
5. Replace the filter drier core with 63-0410.
6. Pressure test and evacuate the circuit as described in the NK-600 Service Manual and according to local regulations.
7. Charge by weight to the nameplate charge of the circuit. If refrigerant was added to this circuit at an earlier visit for an L-03 alarm, do not return the recovered amount blindly: the nameplate charge is the target.
8. Fit the new stepper cable. The rev 2 cable does not fit the rev 3 motor and must not be modified.
9. In Service > Configuration > Circuit 1 > EXV type, select "rev 3". With the wrong setting the valve opens too slowly.
10. Repeat steps 3 to 9 for circuit 2.
11. Start the unit. After 30 minutes check on both circuits that the EXV position follows the request and that superheat is stable.
12. Fit the campaign label next to the nameplate. Log the work and the refrigerant quantities in the unit's F-gas record.
13. Return the two removed valves to Nordkyl within 30 days with the return label from the partner portal. The returned valves are part of the claim.

PARTS

Part no.	Description	Qty per unit
63-1180	EXV-2 expansion valve assembly rev 3, with stepper cable and campaign label	2
63-0410	Filter drier core, NK series	2
63-1170	EXV-2 expansion valve assembly rev 2. Removed parts, return to Nordkyl. Blocked for sale.	(2 returned)

LABOUR ALLOWANCE AND CLAIM

Item	Covered by Nordkyl
Labour	3.5 h per unit, both circuits. Includes diagnosis, recovery, valve and drier replacement, pressure test, evacuation, recharge and start-up.
Travel	One flat rate per unit, per partner agreement
Parts	63-1180 and 63-0410, free of charge
Refrigerant	Make-up refrigerant for losses during the work, up to 1.0 kg per circuit
Warranty status	Covered regardless of the warranty status of the unit

Claim in the Nordkyl partner portal under campaign code NK-C25-14 within 30 days of the work, with: serial number, date, EXV request and position readings from the diagnosis, refrigerant recovered and charged per circuit, and the technician's certificate number. The campaign is open until 30 June 2027. Units not completed by then are handled case by case.

DISTRIBUTION

- Nordkyl authorised service partners, EU
- Nordkyl Service Denmark, Germany, Benelux and Poland
- Nordkyl Product Support
- Nordkyl Sales EU (for information)

REVISION

Rev.	Date	Change
0	2 September 2025	First issue

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